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Evidence for "subsurface" hydrogen on Pd(110): An intermediate between chemisorbed and dissolved species

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We report experimental evidence for the formation of a "subsurface" hydrogen species at 130 K on a Pd(110) single crystal surface. This is well separated by distinct activation energy barriers from species which are chemisorbed or which are dissolved in the Pd bulk, thereby confirming a model of H absorption in Pd in which a subsurface state acts as a reaction intermediate.

I. INTRODUCTION

The interaction of hydrogen with palladium is of considerable importance because of the ability of Pd to form metallic hydrides and thereby to act as a medium for hydrogen storage. Numerous experiments have been carried out using polycrystalline Pd samples and many data are available, e.g., for the activation energy barriers involved in permeation of hydrogen through Pd membranes or in hydrogen diffusion into the bulk as well as for the energetic levels of the respective interstitial sites.[1] However, until now only little progress has been made in relating these data to results obtained from single crystal surfaces.

For a Pd(110) surface exposed to hydrogen at room temperature and higher, Conrad, Ertl, and Latta[2] reported fairly rapid dissolution of hydrogen into the bulk crystal as indicated by thermal desorption spectroscopy. With H/Pd(111) at room temperature, Engel and Kuipers,[3] using a modulated molecular beam technique, likewise found substantial hydrogen penetration into the Pd bulk. They were able to calculate diffusion rate constants and activation energies and described the combined adsorption/absorption phenomena by a simple potential energy diagram. However, at temperatures below 250 K the rate of uptake of the Pd(111) bulk was slow. We observed a rather similar behavior for the Pd(100)/H system,[4] where bulk dissolution effects were found to become appreciable only near and above room temperature. For both the (111) and the (100) face therefore activation energy barriers exist which effectively prevent the hydrogen from diffusing into the bulk at temperatures below 250 K.

The crystallographically more "open" Pd(110) surface behaves quite differently when exposed to hydrogen between 130 and 250 K. As our results will demonstrate there is strong evidence that a "subsurface" hydrogen species is formed just below the topmost Pd atoms, which apparently provides an important link between the chemisorbed surface hydrogen and the dissolved bulk species. An intermediate, "weakly chemisorbed" state was previously suggested by different authors.[5,6] Also, as will be shown later, isotope[7] and contamination effects[1] observed for hydrogen permeation can easily be understood on the basis of this model.

II. EXPERIMENTAL

The experiments were performed in a standard stainless steel system equipped with facilities for LEED, AES, work function ($\Delta\phi$), and thermal desorption (TDS) measurements. Further details concerning apparatus and procedures will be given elsewhere.[8] The residual gas (1.5×10^{-10} Torr) consisted mostly of hydrogen. Coadsorption of CO or H_2O during exposure to hydrogen could be ruled out within the limits of the S/N ratio in the TD spectra.

III. RESULTS AND DISCUSSION

Hydrogen adsorption on Pd(110) resembles that on Ni(110) in terms of the structural properties.[9,10] The two ordered structures formed on Pd(110) at 130 K are also detected on Ni(110)[10,11]: Namely a (2×1) containing a glide symmetry plane and a (1×2) structure at higher coverages. Results of a LEED based structure determination are still pending,[12] but there is good evidence to assume the same geometries as determined by He diffraction on Ni(110): The H atoms are aside the bridges along the dense packed rows in the (2×1), and a metal reconstruction occurs for the (1×2) structure. The latter is indicated also for Pd by the change in the integral order LEED intensity-voltage (IV) curves. An H coverage of $\theta=1$ is associated with the (2×1), and $\theta=1.5$ then corresponds to the (1×2).[8,10] Higher exposures of hydrogen cause the formation of streaks in the [01] direction, thereby pointing to a progressive surface reconstruction. TDS reveals several binding states in which, however, specific differences between Ni(110) and Pd(110) become apparent.

As Fig. 1 demonstrates, the H/Pd(110) TD-spectra exhibit two chemisorbed β states: The β_2 state with its maximim shifting from 350 to 300 K with increasing coverage and at $\theta\geq1$ the β_1 state, growing in as a low temperature shoulder on the β_2 state. At even higher exposures and a coverage exceeding 1.5 monolayers two sharp peaks, denoted as α_1 and α_2, emerge at 160 and 224 K, respectively, which are filled simultaneously. We further note that at an adsorption tem-

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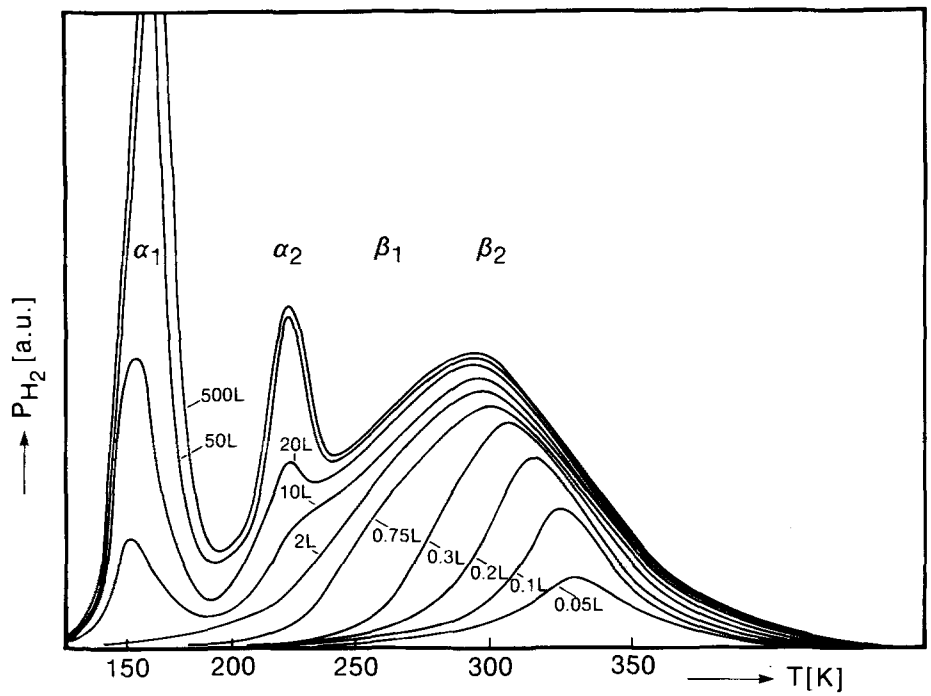


FIG. 1. Series of TD spectra of hydrogen from a Pd(110) surface obtained after adsorption at 130 K.

perature of 130 K, on Pd(110), there is no evidence of any high temperature desorption states ($T_{max} \geq 600$ K), caused by dissolved hydrogen leaving the surface by a diffusion controlled reaction, as is found upon adsorption at room temperature.[2]

The maximum of the narrow α_2 peak is invariant with coverage at 224 K and saturates at about 50 L. It appears to be identical to the one observed by Aldag and Schmidt[14] for desorption from Pd wires following adsorption at 100 K. It is not caused by molecularly absorbed hydrogen, since both α states undergo rapid hydrogen–deuterium exchange. This peak is obviously associated with a structural transition of the Pd surface: A comparison between LEED and TDS shows that the reconstruction is lifted during desorption at this temperature. A similar peak was observed some years ago for H/W(100)[15] and, just recently, for H/Ni(110)[11,13]; both of these surfaces are reconstructed by adsorbed hydrogen. In the latter case also the sharp α_2 peak occurs simultaneously with the removal of the reconstruction, as for Pd(110).

It is remarkable that for H/Pd(110) the α_2 peak only appears at $\theta > 1.5$, whereas the reconstruction already occurs at $\theta > 1$. This can be understood in the model of a subsurface hydrogen: During thermal desorption runs at coverages below $\theta = 1.5$ simultaneous $\Delta\phi$ measurements indicate the loss of hydrogen from the surface at lower temperatures than the onset of desorption.[8] Apparently, part of the hydrogen can move below the surface into the subsurface region, in order to desorb at higher temperatures. Therefore, TD spectra in the range $1 \leq \theta \leq 1.5$ do not show the sharp α_2 peak when the reconstruction is lifted, but the hydrogen desorbs in a broad shoulder (β_1) at slightly higher temperatures than the α_2 peak. This desorption from the subsurface might occur either directly or, presumably, indirectly by replacing H atoms previously being adsorbed on the surface. At $\theta > 1.5$, however, the subsurface region becomes populated during adsorption at 130 K as well. Although most of this "subsurface hydrogen" desorbs in the α_1 state, there is a sufficient amount left to prevent at least part of the surface hydrogen from diffusing into the subsurface during the phase transition. An increasing part of the surface hydrogen has to desorb directly during the structural transition, thereby forming the α_2 peak. As at smaller initial coverages, $1.0 < \theta < 1.5$, the hydrogen remaining in the subsurface desorbs in the β_1 peak as described above.

The second sharp peak, the α_1 state was not observed for Ni(110)/H[11,13] which provides further evidence of the penetration of H atoms to the subsurface sites in Pd(110). Its maximum shifts toward higher temperatures with increasing coverage. A zero order process can be ruled out, since the leading edge of the peaks does not obey the typical logarithmic law, and also their shape is not sufficiently asymmetric. Thus a fractional order ($0 \leq n \leq 1$) can be concluded. A possible explanation for this—after molecular condensation is ruled out by isotope exchange—is that most of the H atoms desorbing from the surface are immediately replaced by atoms moving up from below the surface. The high velocity of this replacement clearly rules out any diffusion-controlled steps suggesting that the replacement H atoms must be located close to the surface. The saturation behavior is another hint of the formation of a distinct subsurface layer of hydrogen atoms; even after a 500 L exposure the amount of H desorbing in the α_1 state equals roughly $\theta = 1.5$, approximately equal to the population of the chemisorbed species.

What may be of importance here is the extent to which the reconstruction occurs: It is not until the streaks appear in LEED that the α states can be observed in

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TDS. Obviously the hydrogen accumulated below the surface is responsible for the continuously increasing disorder of the surface.

As far as energetics are concerned, an evaluation of the leading edge and the peak shift results in an activation energy for desorption of 7 ± 1 kcal/mol, depending on the assumed order. From hydrogenation/dehydrogenation experiments on a thin Pd-foil Auer and Grabke[6] postulated a second, weakly bound chemisorption state with a binding energy of 6.8 kcal/mol Lynch and Flanagan[5] arrived at 8.7–10.5 kcal/mol for a weakly bound state as derived from equilibrium data at infinitely low bulk hydrogen concentrations. Bennet and Fuggle[16] found that the degassing rate of a hydrogen loaded Pd foil between 80 and 400 K is initially limited by a surface process with an activation energy of 8.1 kcal/mol. These energies are in good agreement with our result which may serve as another indication of the subsurface character of this state. A further indication of the nature of the α states is provided by the work function behavior: At $\theta = 1.5$, where the sharp (1×2) LEED pattern is fully developed, $\Delta\phi$ equals 325 mV. Further hydrogen exposure—although filling the α states indicates substantial H uptake—does not lead to a $\Delta\phi$ increase anymore. This behavior can easily be rationalized if the additional H atoms go below the surface and do not contribute to the surface dipole moment.

As mentioned before, isotope exchange experiments show complete exchange for both α states. However, the kinetics of ad/absorption into these states shows pronounced differences for hydrogen and deuterium. Whereas the population of the chemisorbed β states occurs with the same rate—including corrections for differences in ionization probability rate of impingement—the α states are filled much slower by deuterium than by hydrogen. The ratio of the initial filling rates r_i $(\theta_\alpha \to 0)$, as derived from the heights of the TD peaks, is determined to be $r_{H,i} : r_{D,i} = 4.5$ for both peaks at 130 K, which points to a higher barrier for deuterium than for hydrogen to populate the α states on the Pd(110) surface. This result contrasts the well-known "inverse isotope effect" in hydrogen/deuterium bulk diffusion,[1] where the activation energy for diffusion is about 0.5 kcal/mol lower for D atoms than for H atoms. It has to be noted that these principle differences between bulk absorption and surface "sorption" isotope effects render an experimental distinction of these two processes possible.

The activation energy for hydrogenation of 6.8 kcal/mol mentioned before can not refer to the barrier between the chemisorbed and the subsurface state but rather to the one between subsurface and dissolved hydrogen states. Activated adsorption into the subsurface state with a 6.8 kcal/mol barrier would imply a very low sticking probability ($\leq 10^{-5}$) at 130 K for the α states which was not observed experimentally.[8] In fact, during the Pd phase transition $[(1\times2) \rightarrow (1\times1)]$ the barrier from hydrogen adsorbed on the (1×2) reconstructed surface to the subsurface region must be lower than that to desorb: Heating that surface causes part

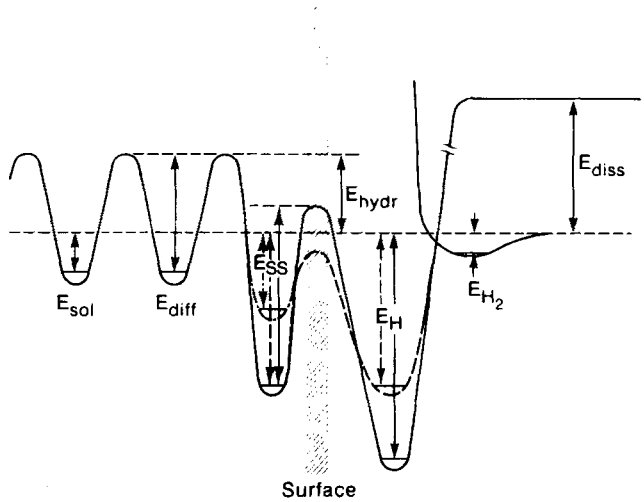


FIG. 2. Potential energy diagram for the adsorption, surface sorption into the subsurface states and bulk absorption for the system H/Pd(110).

	E (kcal/mol)
E_{sol} = heat of solution (Ref. 1)	4.6
E_{diff} = activation energy for bulk diffusion (Ref. 1)	5.5
E_{hydr} = activation energy for hydrogenation (Ref. 6)	6.8
E_{ss} = activation energy for desorption from the H subsurface state	15 (rec., $> \theta$: 7)
E_H = activation energy for desorption from the chemisorbed state	20 (rec.: 12)
E_{H2} = binding energy of a (physisorbed) molecule	...
E_{diss} = heat of dissociation of a hydrogen molecule	104 .

Full line: Unreconstructed surface; dashed line: Reconstructed surface, < subsurface H concentration; dotted line: Reconstructed surface, > subsurface H concentration.

of the hydrogen to dissolve into the subsurface region rather than to desorb. The barrier for hydrogen adsorbed on the nonreconstructed surface to the subsurface region is apparently higher, since at $\theta < 1$ only a small amount of hydrogen dissolves into that region upon heating, as indicated by $\Delta\phi$ measurements. Finally, the net activation energy for the α_1 state to desorb must be lower than that to move into the bulk, since addition of thermal energy forces the hydrogen to preferentially desorb and not to diffuse into the bulk.

On the basis of the above data and utilizing literature data to describe the bulk hydrogen states, a potential energy diagram can be constructed which is reproduced in Fig. 2. It exhibits the following features: Chemisorption on the nonreconstructed surface (H_{ad}, full line) including a molecular precursor (H_2) is indicated by the full line. This part is identical to the diagram reported by Engel and Kuipers[3] for H/Pd(111). A second chemisorption state, marked by the dashed line, represents hydrogen adsorbed on the reconstructed surface (H_{ad}, dashed line). As described above, part of the adsorbed hydrogen moves into the subsurface region during warming up to $T \geq 200$ K. The dashed line displays an energy barrier to the subsurface state (H_{ss})

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lower than that to desorb, thus this process is not activated with respect to the vacuum level. Penetration into that region on the nonreconstructed surface (full line) might be activated, the activation energy is not known. The energy level of the subsurface state under the reconstructed surface E_{ss} appears to be coverage dependent, with higher concentration of hydrogen an increasing amount is on a higher energy level (dotted line), responsible for the α_1 peak in TDS. The position of this level with respect to the vacuum level can be directly extracted from the TDS data, since its population is a nonactivated process. During desorption the Pd phase transition takes place, the remaining part of the subsurface hydrogen desorbing in the β_1 peak. The related state is represented again by the full line. Its absolute level is not known, since from TDS only the activation energy for desorption is evaluated which, in the case of activated adsorption with an unknown activation barrier gives no information concerning the absolute level of the state. This tentative energy diagram emphasizes the importance of the H induced surface reconstruction for the formation of the subsurface hydrogen (tentative, since the possibility of an even more complicated "true" energy diagram cannot be ruled out). The coverage dependence of the different states is as follows: For the subsurface hydrogen under the reconstructed surface it is already described above, whereas under the nonreconstructed surface no major coverage effect appears, presumably because it is limited to a low concentration. The hydrogen adsorbed on either of the surfaces is only slightly effected by the H–H interactions.[8] It is not possible to give any estimate of how higher concentrations of subsurface hydrogen affect the binding energy of the adsorbed hydrogen, since this constellation is not accessible with TDS.

The isotope effect observed for populating the subsurface is not included in this diagram, since its origin is not obvious. A possible reason involves the different vibrational energy contributions for the activation energies to either desorb or migrate from the reconstructed surface into the subsurface. This would imply that a cluster with locally high coverage ($\theta > 1.5$) of hydrogen adsorbed on the reconstructed surface acts as a transition state for the subsurface population.

For H/Pd(100), where likewise no α peaks were detected but bulk diffusion occurred at $T > 300$ K (cf. TD spectra of Ref. 4), Comsa *et al.*[17] proposed a potential diagram for which they calculated a barrier height of 3.15 kcal/mol for D atoms desorbing directly from below the surface. This value, obtained from a permeation experiment, is almost identical with the barrier for deuterium migration into the bulk—3.3 kcal/mol.[1] The existence of a distinct subsurface state therefore can be excluded, which contrasts recent findings by Eberhardt, Greuter, and Plummer,[18] who proposed a similar species for the (111) faces of Pt, Pd, and Ni on the basis of UV photoemission data. Serious doubt was raised about their interpretation, however, at least for Pd(111) by recent work on D_2 adsorption on Pt(111).[19]

The question arises as to what extent the H induced reconstruction governs the formation of the subsurface

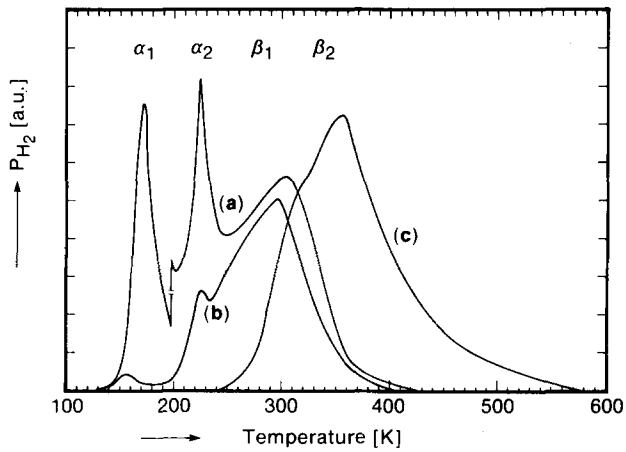


FIG. 3. TD spectra after 100 L hydrogen exposure at 130 K to (a) clean surface, scale for α_1 state: *0.5; (b) surface preexposed to 0.4 L CO; (c) clean surface and subsequent exposure to 10 L CO.

species. It has to be kept in mind that these states are only filled after the chemisorption sites on the surface have been saturated, and the reconstruction has taken place. Any adsorbate that influences the reconstruction should therefore have a marked effect on the population of the α states. Adsorbed CO, for example is found to not only block Pd surfaces for H adsorption,[1,20] but also to inhibit H passage through the surface in both directions: Into the bulk and from the bulk into the gas phase, as found for permeation through a Pd foil.[1] In CO and H coadsorption experiments we found that even fairly small coverages of CO on Pd(110) impare the population of the α states, whereas the chemisorbed β states are much less affected, as shown in Fig. 3. Thus one can conclude that chemisorbed CO increases the energy barrier between the chemisorbed and subsurface hydrogen states, in addition to geometrically blocking hydrogen adsorption sites. Simultaneously, the hydrogen induced surface reconstruction is inhibited. At higher CO coverages the population of the chemisorbed hydrogen decreases too, up to a complete poisoning.

The effect of adsorbing CO on preadsorbed hydrogen is similar. In addition the CO apparently forces the hydrogen below the surface, an effect already assumed by Bliznakov and Kiskinova[21] for coadsorption at room temperature on a Pd wire. On Pd(110), after preadsorption of sufficient hydrogen to populate the α states, subsequently adsorbed CO lifts the reconstruction and finally a (2×1) pattern characteristic of CO is formed. In the TD spectra, not only the α peaks are completely suppressed, but also the β peaks are affected. H desorption occurs in one broad peak only after the CO starts to desorb, with a CO peak temperature of 470 K, comparable to that on a clean surface. The increase of the energy barrier between chemisorbed and subsurface hydrogen states is obviously affecting the "transport" between these states in both directions.

IV. CONCLUSIONS

In summary, during hydrogen adsorption on Pd(110) initially the usual chemisorption state is filled. At

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coverages $\theta \geq 1$ the surface reconstructs, now having a local coverage of $\theta = 1.5$, again in a chemisorbed state. In contrast to the other low index Pd planes, a distinct state exists for subsurface hydrogen, different from hydrogen dissolved in the bulk. The barrier between surface and subsurface states is lower than the activation energy for bulk diffusion, presumably due to the reconstruction of the surface. This is why this site can slowly be populated at 130 K, after the surface is reconstructed and saturated with adsorbed hydrogen. Upon heating this subsurface hydrogen desorbs rapidly, the rate of desorption not being controlled by diffusion processes. Only at room temperature (and higher) hydrogen can also diffuse into deeper lying bulk regions, which is kinetically limited at low temperatures. The rate of population of the subsurface region exhibits an isotope effect, different from the inverse isotope effect for bulk diffusion; it also is strongly diminished by coadsorbed CO.

The different characteristic features during hydrogen adsorption and desorption can be explained by the energy diagram proposed in Fig. 2: The differences in potential energy for H adsorbed on the reconstructed Pd surface compared to on the nonreconstructed one are mainly responsible for the complex behavior.

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[1]E. Wicke and H. Brodowsky in *Hydrogen in Metals*, edited by G. Alefeld and J. Voelkl (Springer, Berlin, 1978).
[2]H. Conrad, G. Ertl, and E. E. Latta, Surf. Sci. **41**, 435 (1974).
[3]T. Engel and H. Kuipers, Surf. Sci. **90**, 162 (1979).
[4]R. J. Behm, K. Christmann, and G. Ertl, Surf. Sci. **99**, 320 (1980).
[5]J. F. Lynch and T. B. Flanagan, J. Phys. Chem. **77**, 2628 (1973).
[6]W. Auer and H. J. Grabke, Ber. Bunsenges. **78**, 58 (1974).
[7](a) G. Bohmholdt and E. Wicke, Z. Phys. Chem. NF **56**, 133 (1973); (b) G. Hollek and E. Wicke, *ibid.* **56**, 155 (1973).
[8]M.-G. Cattania, V. Penka, R. J. Behm, K. Christmann, and G. Ertl, Surf. Sci. (to be published).
[9]T. N. Taylor and P. J. Estrup, J. Vac. Sci. Technol. **11**, 244 (1974).
[10]T. Engel and K. H. Rieder, Surf. Sci. **109**, 140 (1981).
[11]F. Chebab, K. Christmann, G. Ertl, and V. Penka (in preparation).
[12]W. Moritz, V. Penka, R. J. Behm, M.-G. Cattania, K. Christmann, and G. Ertl (in preparation).
[13]A. Winkler and K. D. Rendulic, Surf. Sci. **118**, 19 (1982).
[14]A. W. Aldag and L. D. Schmidt, J. Catal. **22**, 260 (1971).
[15]P. W. Tamm and L. D. Schmidt, J. Phys. Chem. **54**, 4775 (1971).
[16]P. A. Bennet and J. C. Fuggle (to be published).
[17]G. Comsa, R. David, and B.-J. Schumacher, Surf. Sci. **95**, L210 (1980).
[18]W. Eberhardt, F. Greuter, and E. W. Plummer, Phys. Rev. Lett. **46**, 1085 (1981).
[19]P. R. Norton, J. A. Davis, and T. E. Jackman, Surf. Sci. **121**, 103 (1982).
[20]H. Conrad, G. Ertl, and E. E. Latta, J. Catal. **35**, 363 (1974).
[21]G. Bliznakov and M. Kiskinona, Proceedings of the 4th International Symposium on Heterogeneous Catalysis, Varna, 1979, p. 39.

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